

Section 3.2 Exercises

Directions: For exercises 1 – 8, rewrite the quadratic functions in standard form and give the vertex.

1. $f(x) = x^2 - 12x + 32$

2. $g(x) = x^2 + 2x - 3$

3. $f(x) = x^2 - x$

4. $f(x) = x^2 + 5x - 2$

5. $h(x) = 2x^2 + 8x - 10$

6. $k(x) = 3x^2 - 6x - 9$

7. $f(x) = 2x^2 - 6x$

8. $f(x) = 3x^2 - 5x - 1$

Directions: For exercises 9 - 15, determine whether there is a minimum or maximum value to each quadratic function. Find the value and the axis of symmetry.

9. $y(x) = 2x^2 + 10x + 12$

10. $f(x) = 2x^2 - 10x + 4$

11. $f(x) = -x^2 + 4x + 3$

12. $f(x) = 4x^2 + x - 1$

13. $h(t) = -4t^2 + 6t - 1$

14. $f(x) = \frac{1}{2}x^2 + 3x + 1$

15. $f(x) = -\frac{1}{3}x^2 - 2x + 3$

Directions: For exercises 16 - 20, determine the domain and range of the quadratic function.

16. $f(x) = (x - 3)^2 + 2$

17. $f(x) = -2(x + 3)^2 - 6$

18. $f(x) = x^2 + 6x + 4$

19. $f(x) = 2x^2 - 4x + 2$

20. $k(x) = 3x^2 - 6x - 9$

Directions: For exercises 21 - 26, sketch a graph of the quadratic function and give the vertex, axis of symmetry, and intercepts.

21. $f(x) = x^2 - 2x$

22. $f(x) = x^2 - 6x - 1$

23. $f(x) = x^2 - 5x - 6$

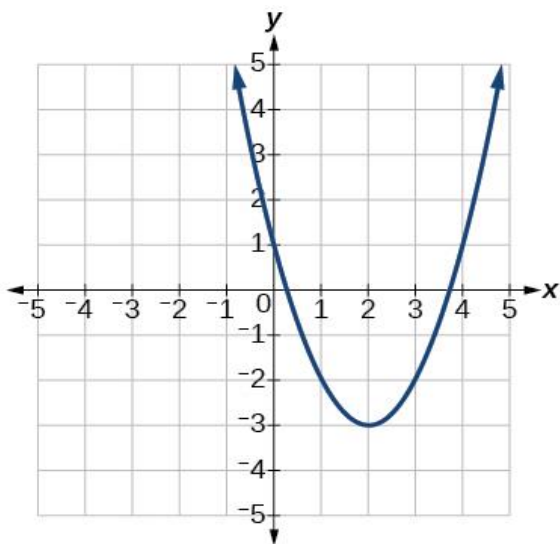
24. $f(x) = x^2 - 7x + 3$

25. $f(x) = -2x^2 + 5x - 8$

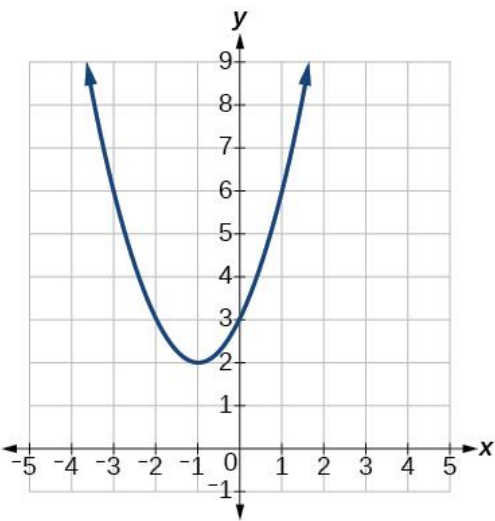
26. $f(x) = 4x^2 - 12x - 3$

Directions: For exercises 27 - 32, write the equation for the graphed function.

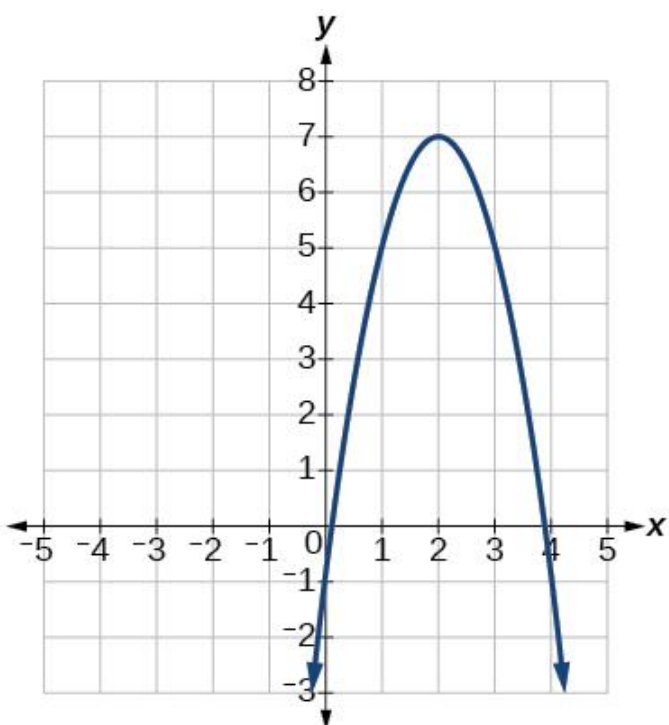
27.



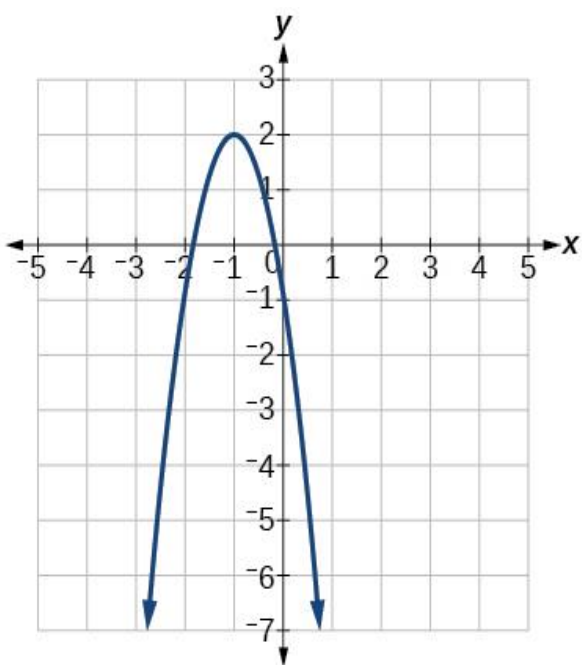
28.



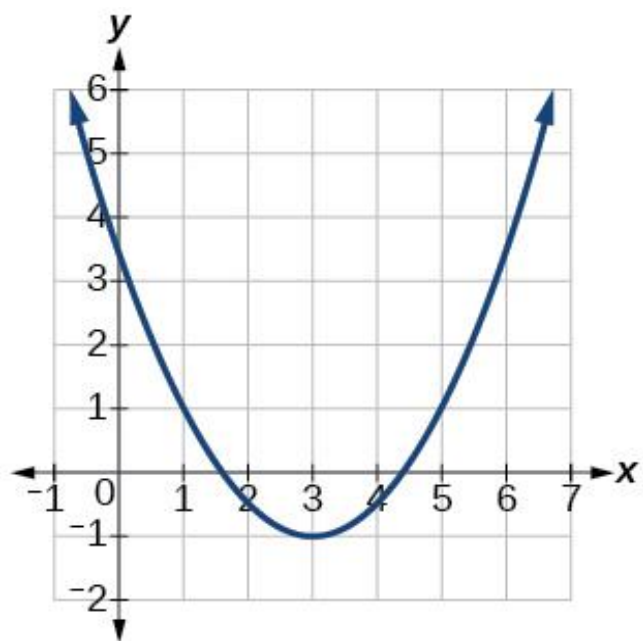
29.



30.



31.



32.

